

CIND820: Final Report

Project Name: Predicting Success of Video Games

Student: Eddie Li

Student #: 500820042

Supervisor: Ceni Babaoglu

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1 Research Questions

This study investigates some factors that could influence global video game sales and evaluates their influence on predicting those sales. The following research questions were used as a guideline for the study:

- Among platform, genre, review score, which features are most influential in predicting global sales?
 - Understanding which features are important helps to quantify each variable’s impact and allows accurate sales forecasting [1][2].
- Can global video game sales be predicted using factors such as platform, genre, review score, and release year?
 - Examining predictive accuracy addresses the challenge posed by complex interactions among game features and consumer behavior [3].
- Are certain platforms or genres inherently more predictable in terms of their sales outcomes than others?
 - Investigating predictability across genres and platforms will determine whether predictive models perform equally well on each [4].

2 Contributions of Study

This study contributes to the existing literature on video game sales prediction in a few ways. The first way it contributes is by integrating multiple factors of sales into a model. While the majority of models from earlier studies often considered factors such as genre, platform, or reviews independently, this study evaluates their combined impact on global sales [5].

Second, the study extends the existing literature by applying and evaluating nonlinear machine learning models. Previous studies have shown that relationships between factors such as review scores and sales are often nonlinear. Even so, many studies rely on

linear approaches. By implementing models such as Random Forest, Gradient Boosting, K-Nearest Neighbors (KNN), Artificial Neural Networks (ANN), and Support Vector Regression (SVR), this study can capture more complex patterns in the data [1].

Lastly, this project contributes via a thorough model evaluation with both a regular train-test split validation and a 5-fold cross validation. This improves the reliability of model evaluation and highlights generalization issues that are often overlooked [6].

3 Literature Review

Prior literature identifies several factors of video game sales which include factors like game characteristics, reputation, and platform selection. Game characteristics such as genre, themes, and release timing play a large role in global sales [2]. Previous studies show that certain genres are more likely to achieve higher sales, while market conditions and competition at the time of release can also influence performance [2].

Reputation related factors are also widely recognized as an important factor of sales. Both consumer and critic reviews act as indicators of quality and can influence purchasing decisions [7]. Prior research shows that reviews have a stronger impact on less popular titles since they do not have an established audience. This effect on sales for less popular titles is often nonlinear and depending on how extreme the review scores are. In addition, publisher and franchise reputation contribute to commercial success since established brands tend to attract more consumers [8].

Platform selection is another important factor identified in the literature. Previous literature showed that the same game can perform differently across platforms due to variations in platform demographics and regional preferences [9][4]. Platform competition and exclusivity deals/strategies further influence global sales by affecting accessibility and brand reputation [10].

In addition to identifying key factors, prior research emphasizes the importance of predictive modeling techniques. Data preprocessing, including handling missing values and preparing variables, has been shown to improve predictive accuracy [3]. Exploratory

analysis of historical data helps identify relationships between variables and supports feature selection for modeling [1]. Regression-based approaches are commonly used in this domain, with studies showing that nonlinear models often outperform linear models due to the complex relationships between variables [11].

Despite these advancements, gaps remain in the literature. Many studies examine determining factors in isolation rather than analyzing their combined effects [1]. Furthermore, traditional modeling approaches often rely on linear methods, which may not fully capture nonlinear relationships between variables such as review scores and sales [11].

4 Integration of Literature in Study

Prior research identified all factors in the selected data set as determining factors of video game sales. This helped guide the selection of variables used in the predictive models [2][9][12]. These factors were analyzed alongside one another using nonlinear methods to identify their combined impact. This will help address the gap in the literature where variables are often examined independently rather than collectively [5]. In addition, findings from previous studies highlighting the importance of data preprocessing and exploratory analysis influenced the implementation of techniques such as handling missing values, encoding categorical variables, and standardizing numerical features. These steps ensured that the dataset was properly prepared for analysis and modeling [7].

The literature that focused on the importance of nonlinear relationships between factors largely shaped the modeling approach. Prior studies demonstrated that relationships between factors such as review scores and sales are often nonlinear and that nonlinear models tend to outperform linear approaches in predictive tasks [1][3][13]. Therefore, rather than relying on linear regression methods, this study implemented multiple machine learning models to better capture complex patterns. The machine learning models include which include: Random Forest, Gradient Boosting, K-Nearest Neighbors (KNN), Artificial Neural Networks (ANN), and Support Vector Regression (SVR). The use of train-test splitting and 5-fold cross-validation was also informed by established best prac-

tices in predictive modeling to ensuring a more reliable evaluation of model performance [6][11].

5 Methodology and Study Design

The methodology consists of preprocessing, exploratory data analysis, model development, and performance evaluation. The dataset used in this study was obtained from a publicly available website called 'Kaggle' from the following website url [14]: "https://www.kaggle.com/datasets/thedevastator/global-video-game-sales-and-reviews"

The dataset includes variables: such as platform, genre, release year, and review scores. These variables are used as input variables to predict the target variable: global sales.

Data preprocessing steps were applied to ensure the dataset was suitable for modeling. The steps included:

- Removing columns that contain a missing "Year" value
- Converting year from float64 to int64
- Converting object data types to string
- Rounding the float64 columns to 2 decimal places
- Converting categorical variables (Platform and Genre) into numerical format using one-hot encoding
- Standardizing numerical features (Review Score and Year) using StandardScaler

These preprocessing steps are consistent with established practices in predictive modeling [7]. Exploratory data analysis (EDA) was also performed to identify patterns, trends, and relationships between variables, supporting feature selection and model design. The results of the exploratory data analysis is included in the findings and interpretation section.

After preprocessing and exploratory data analysis, multiple multiple nonlinear models were developed to estimate global video game sales. Nonlinear machine learning models were used since many previous studies relied on linear methods despite evidence showing that relationships between variables such as review scores, platform, and sales are often nonlinear [1][3][13].

The nonlinear models used in this study and the reasoning are included below [15][16]:

- Random Forest Regression - Combines many trees to handle nonlinear relationships between features.
- Gradient Boosting Regression - Builds trees step by step and fixes mistakes along the way to improve accuracy.
- K-Nearest Neighbour (KNN) - Predicts outcomes based on similar past examples.
- Artificial Neural Network (ANN) - Can approximate highly nonlinear functions and learn complex feature interactions simultaneously.
- Support Vector Regression (SVR) - Uses feature mapping techniques to capture nonlinear relationships in the data.

By developing and comparing multiple nonlinear models, the study aims to determine which approach provides the most accurate and reliable prediction for global video game sales.

Following model selection, the dataset was divided into training and testing subsets using a train-test split methodology. The training set was used to fit the models and the testing set was used to evaluate how well the models would generalize to new data. In addition, 5-fold cross-validation was applied during training to improve the accuracy of the models and reduce the risk of overfitting. Cross-validation is a well-established method for obtaining more accurate and robust estimates of model performance [6].

Model evaluation was conducted using the following metrics:

- R^2 (Coefficient of Determination) - Measures the proportion of variance. Values closer to 1 indicate a better model and negative values indicate that the model

performs worse than simply predicting the mean of the target variable.

- Model Run Time -
- RMSE (Root Mean Squared Error) - Measures the average magnitude of prediction errors and penalizes larger errors more heavily due to the squaring.
- MAE (Mean Absolute Error) - Measures the average absolute difference between predicted and actual values.

These metrics allow for consistent comparison across models and allows for identification of the best performing model.

6 Findings, Interpretation, and Model Evaluation

6.1 Exploratory Data Analysis Trends

Analysis of the dataset and visualizations shows a few key patterns. By looking at *Figure 1*, it can be seen that regional sales across North America, Europe, Japan, and the Rest of World are all strongly positively correlated, which is expected since global sales are the sum of these regions. *Figure 2* indicate that review score are not strongly correlated with global sales because it shows a high concentration of high-scoring games with relatively low sales in the bottom-right region which suggests that high review scores do not necessarily translate to high commercial performance. While *Figure 1* and *Figure 2* agree with each other, *Figure 3* seems to disagree with *Figure 1*. While the video game industry has grown over time with more units sold in recent years and with people buying more games as the years go on [1][17], the correlation with year and global sales is much weaker than expected.

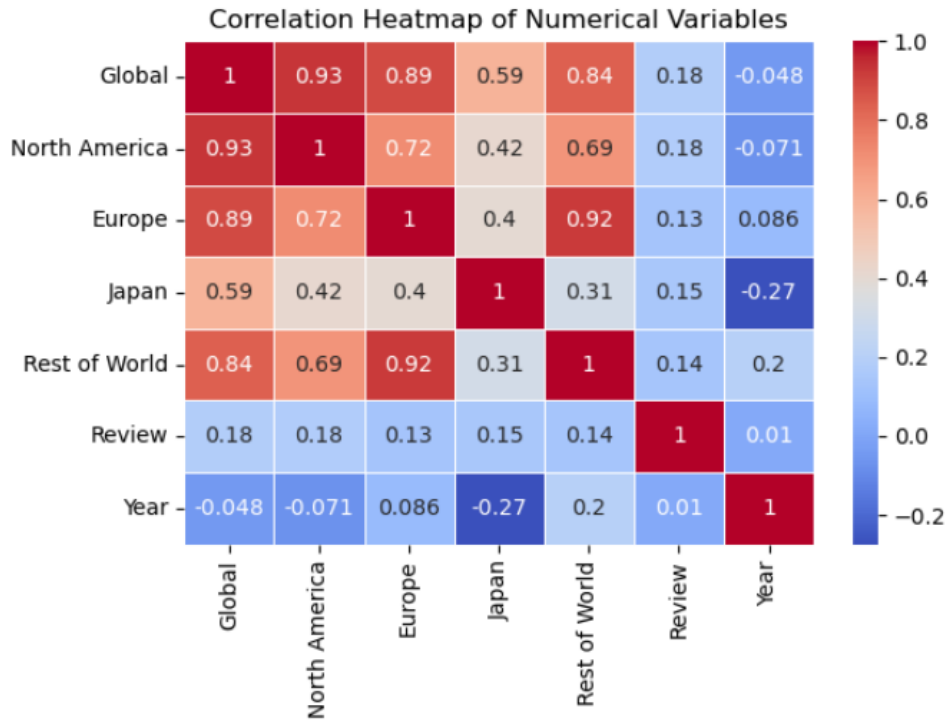


Figure 1: Correlation Heatmap



Figure 2: Log(Global Sales vs Review Score)

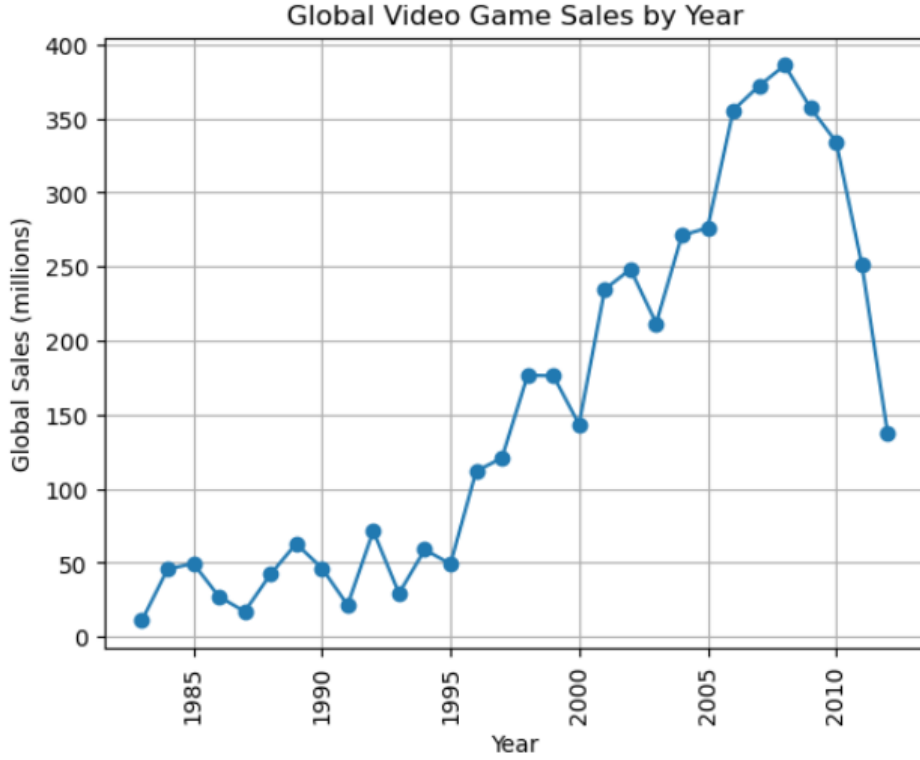


Figure 3: Global Sales vs Year

The visualizations grouped by genre and platform highlight a few notable outliers in the dataset. The median sales are skewed as seen in *Figure 6*, *Figure 10*, and *Figure 11*. Most games sold have similar sales performance when all regions are grouped by genre and platform. Meanwhile, Japan has drastically difference sales performance when it comes to games sold by platform and genre and the Rest of the World when it comes to platform. This indicates that developers/publishers that plan to market to a Japanese market or a Japanese developer/publisher marketing outside of Japan should do extra market research on their target audience [18]. When it comes to marketing to the rest of the world it likely depends on what console is popular in that region and what legislation causes specific games to become region locked [19].

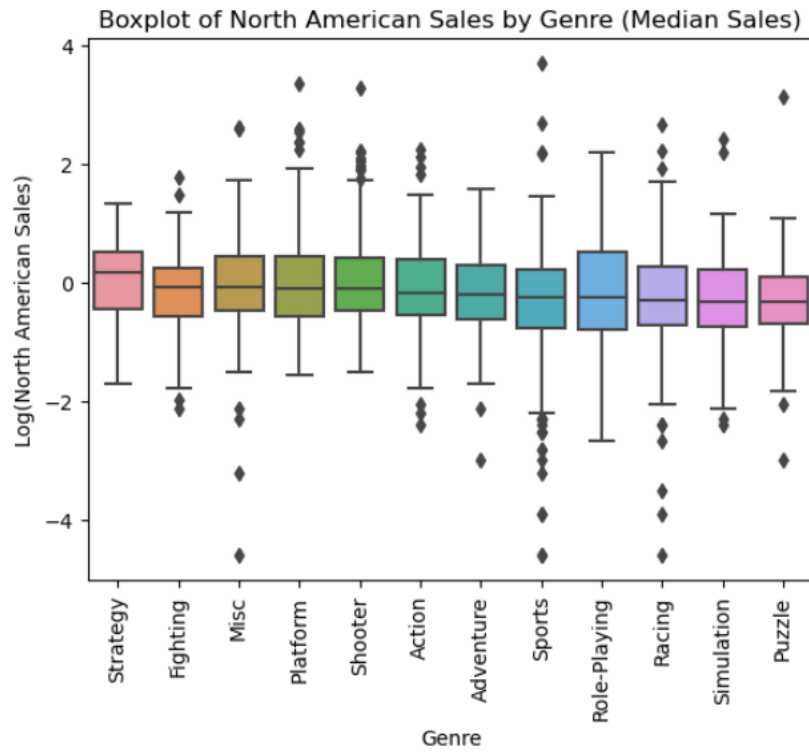


Figure 4: North American Sales Boxplot by Genre

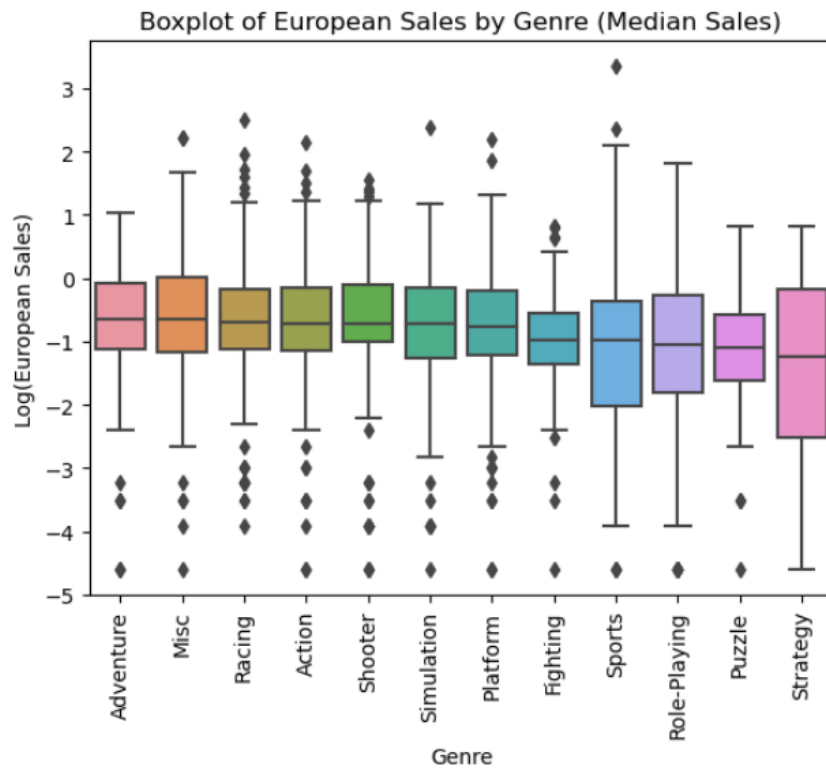


Figure 5: European Sales Boxplot by Genre

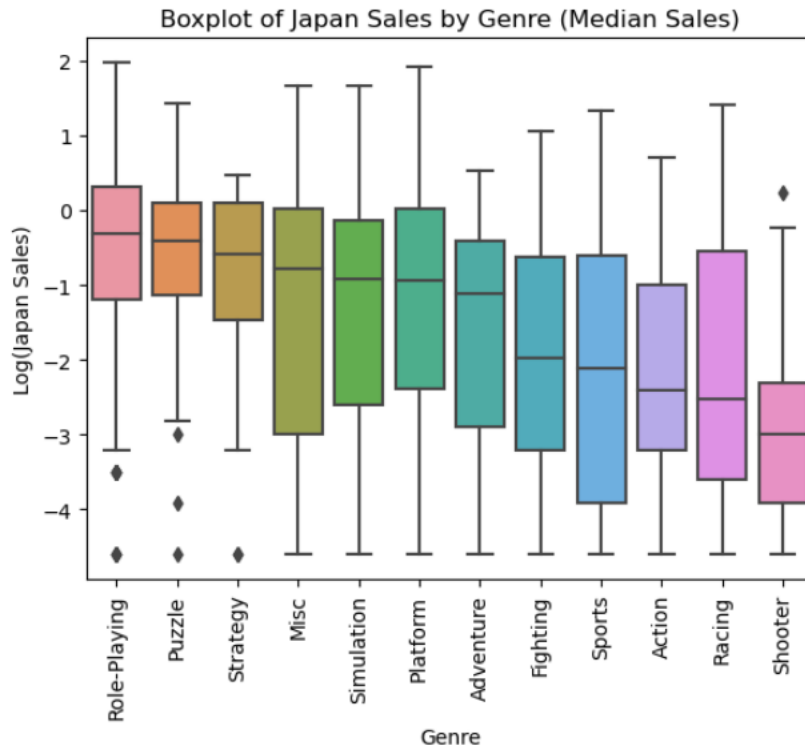


Figure 6: Japanese Sales Boxplot by Genre

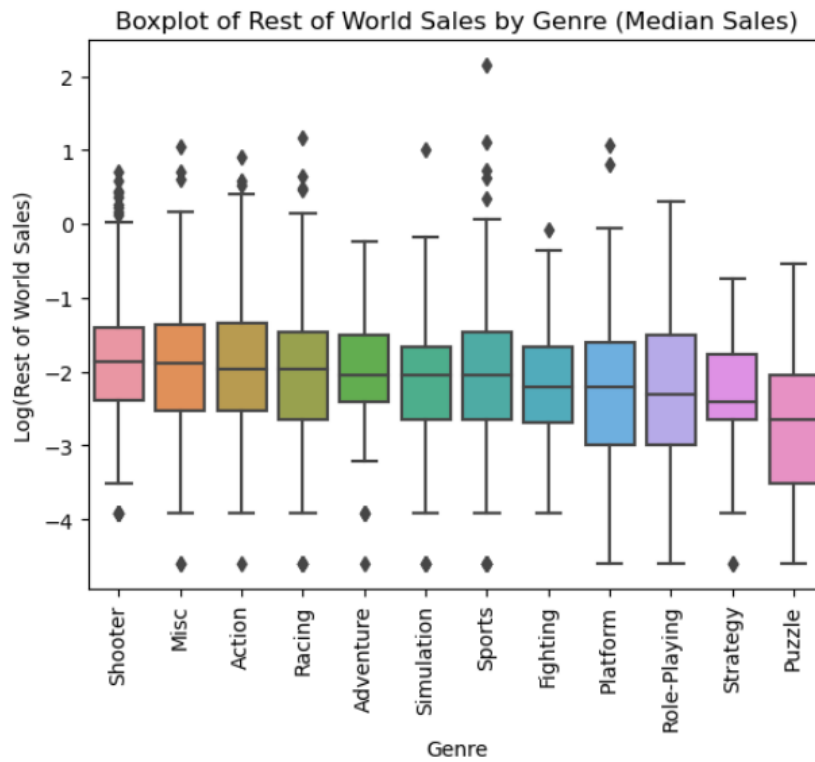


Figure 7: Rest of World Sales Boxplot by Genre

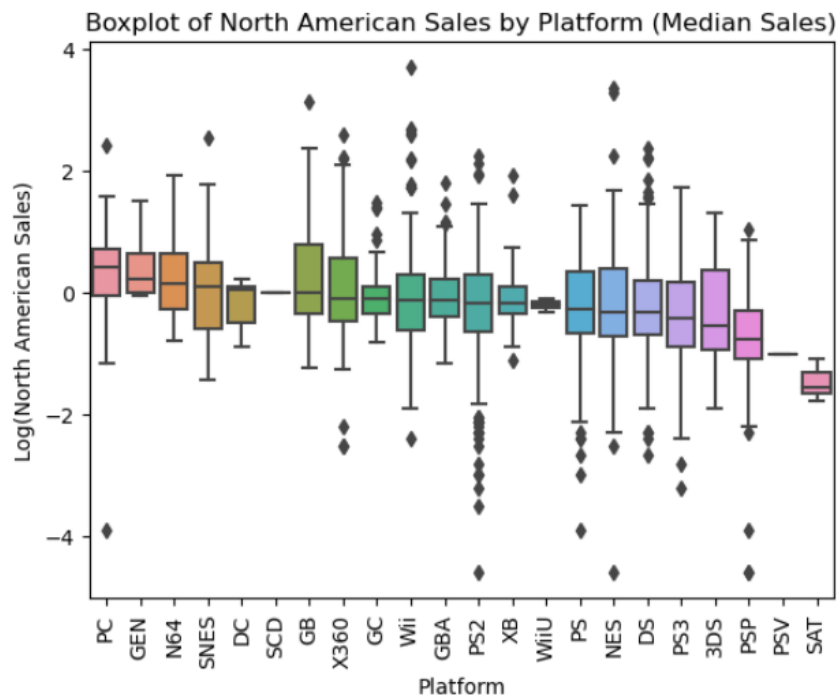


Figure 8: North American Sales Boxplot by Platform

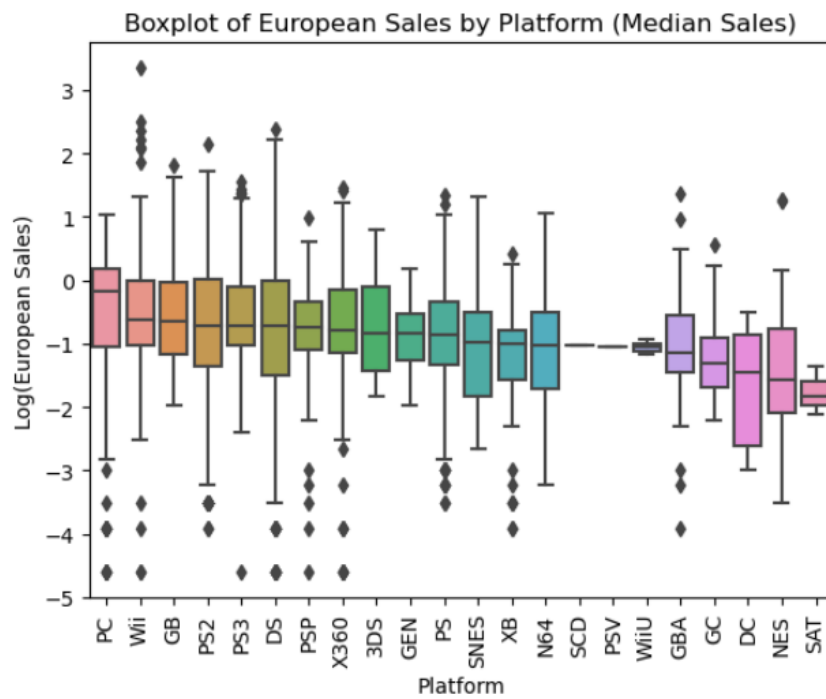


Figure 9: European Sales Boxplot by Platform

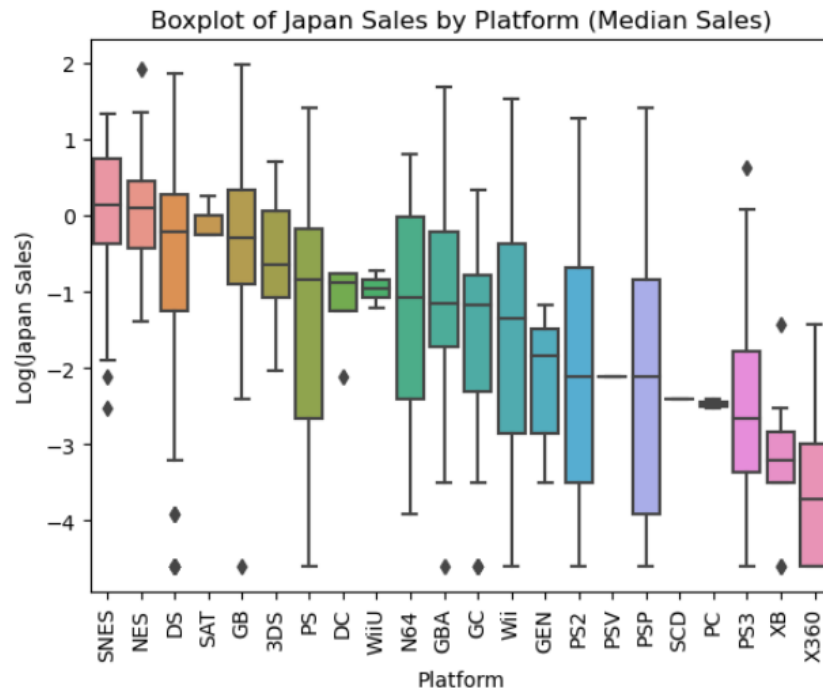


Figure 10: Japanese Sales Boxplot by Platform

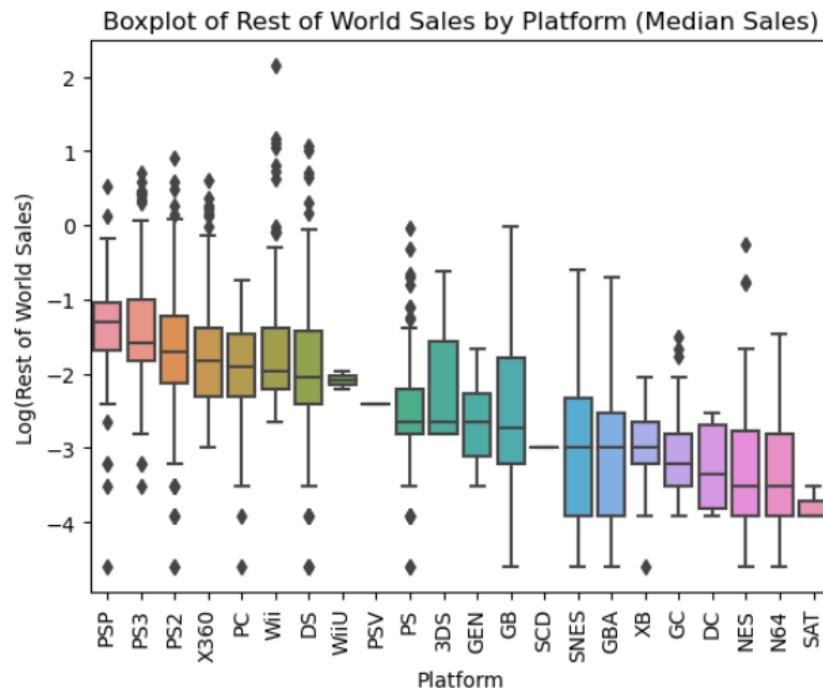


Figure 11: Rest of World Sales Boxplot by Platform

Table 1: Performance Metrics for Regular Train-Test Split of Nonlinear Machine Learning Models

Model	R^2	RMSE	MAE	Training Time (s)	Prediction Time (s)
Random Forest	0.1841	3.0729	1.5967	0.5730	0.0143
Gradient Boosting	0.1003	3.2268	1.6366	0.2186	0.0010
KNN	0.1071	3.2145	1.6477	0.0010	0.1817
ANN	0.2158	3.0126	1.6605	1.6030	0.0000
SVR	-0.0109	3.4204	1.3777	0.0772	0.0300

Table 2: Performance Metrics for Cross-Validation of Nonlinear Machine Learning Models

Model	R^2	RMSE	MAE	Cross-Validation Time (s)
Random Forest	-0.0459	3.4398	1.5423	2.8933
Gradient Boosting	-0.0802	3.5100	1.5554	1.1062
KNN	-0.0310	3.4792	1.5875	0.0440
ANN	-0.2393	3.7540	1.7551	8.0023
SVR	-0.0098	3.4747	1.3176	0.5714

6.2 Effectiveness

The effectiveness of the machine learning models was evaluated based on their ability to accurately predict global video game sales. Therefore, effectiveness was measured using R^2 values [11][15].

In the regular train-test evaluation, R^2 values indicate that Artificial Neural Network (ANN) had the highest predictive accuracy where $R^2 = 0.2158$. Random Forest, Gradient Boosting, and K-Nearest Neighbor all had moderate performance where R^2 was approximately 0.1. Support Vector Regression (SVR) performed the worst, being the only machine learning model to have a negative R^2 value. These results show that Artificial Neural Network (ANN) captured trends in the training data a bit better than the other models and Support Vector Regression (SVR) had the worst fit. These results suggest that while some nonlinear models can capture patterns in the data, their overall explanatory power remains limited.

During 5-fold cross-validation, all models had negative R^2 values. This change from train-test only shows that the machine learning models could not maintain performance across different data subsets. Support Vector Regression (SVR) performed slightly better than other models since it had the smallest negative R^2 value at -0.0098. Comparatively, Artificial Neural Network (ANN) went from the best performing to the worst performing

with an R^2 value of -0.2393.

6.3 Efficiency

Efficiency was evaluated by comparing the computational time required to train and test each model. This comparison helped assess the trade-off between predictive performance and computational cost which checks that the selected model is practical to implement on top of being accurate [11][15].

For the regular train-test evaluation, Artificial Neural Network (ANN) required the most training time 1.6030s. Random Forest and Gradient Boosting had moderate to low computation times of 0.5730s and 0.2186s respectively. The most efficient models were K-Nearest Neighbor (KNN) and Support Vector Regression (SVR) with much faster times of 0.0010s and 0.0772s respectively. Prediction times were low for most models except K-Nearest Neighbor (KNN), the model with the fastest training time. In addition, while Artificial Neural Network (ANN) required more training time, there was no prediction time required for this data set.

Cross-validation results further emphasize computational differences. Artificial Neural Network (ANN) remained the highest in computation time 8.0023s. Both K-Nearest Neighbor (KNN) and Support Vector Regression (SVR) remained the most efficient machine learning models with total times of 0.0440s and 0.5714s respectively. Random Forest and Gradient Boosting showed moderate computation times 2.8933s and 1.1062s respectively, maintaining the same relative position in ranking.

6.4 Stability

Stability assesses how consistently the models perform across different subsets of data. Cross-validation was performed to measure the stability/robustness of the models. [11][15]. While all models had negative R^2 values, the RMSE and MAE values remained fairly consistent across folds. This indicates that the magnitude of predictive error was stable even when the model accuracy struggled. It also indicates that the models are reliable in capturing general patterns in the dataset require improvements to increase

predictive capabilities [11][6].

6.5 Model Ranking and Recommendation

Table 3: Model Ranking Based on Cross-Validation Performance

Model	Effectiveness Rank	Efficiency Rank	Stability Rank
Random Forest	2	4	2
Gradient Boosting	4	3	3
KNN	3	1	4
ANN	5	5	5
SVR	1	2	1

Support Vector Regression (SVR) and K-Nearest Neighbors (KNN) are recommended as the most suitable models based on the evaluation results. Support Vector Regression (SVR) performed the best overall during cross-validation. It had the highest R^2 , the second lowest RMSE, and the lowest MAE. This indicates that it was the most reliable in capturing patterns in the data and maintaining consistent performance across different subsets. K-Nearest Neighbor (KNN) provided a stronger balance between performance and efficiency. It is the third most effective but has the fastest computation time. Although its stability is worse than Support Vector Regression, it still produces quick and effective results. These models represent the best choices for this dataset, with Support Vector Regression (SVR) prioritizing accuracy and stability, and K-Nearest Neighbor (KNN) offering efficiency with solid predictive performance.

7 Limitations and Ethical Considerations

7.1 Limitations

This study has several limitations related to data quality, model assumptions, and computational constraints. First, the dataset used in this project contains includes a few missing values and a limited number of variables. While preprocessing techniques such as handling missing values and encoding categorical variables were applied these steps may not fully fix data quality limitations. Additionally, the dataset lacks potentially

important variables which could include but is not limited to [20]:

- Budgets - Marketing budgets, development budgets, etc.
- Player engagement metrics - Average playtime, concurrent players, average play session duration, player retention rate, etc.
- Pricing and Discounts - The cost of the game, any DLC, and any microtransactions it includes. Discounts affect all these costs.
- Availability - Regional and localizations restrictions that prevent games from releasing globally.
- Exclusivity - Temporary or permanent exclusivity rights to titles.

All these missing variables could increase the accuracy and robustness of the machine learning models. Limitations in data quality and completeness are recognized as factors that can reduce the reliability and predictive performance [21].

Second, the machine learning models in this study depend on specific assumptions and design choices that influence performance. Nonlinear models such as Support Vector Regression (SVR), K-Nearest Neighbors (KNN), Random Forest, Gradient Boosting, and Artificial Neural Networks were chosen to capture complex patterns, but they still rely on factors like parameter choices and data structure [15][16]. If these conditions are not met then model accuracy may be reduced. Overall, proper model selection and configuration are crucial for reliable predictions [22].

Third, computational limits affected this study. While several models were tested, more advanced approaches like deep learning or extensive hyperparameter tuning were not explored due to constraints. While cross-validation was explored to help with model stability, the addition adds to computational cost especially for complex models such as Artificial Neural Networks (ANN). Balancing model complexity, computation, and performance is a common challenge in machine learning [22].

7.2 Ethical Considerations

In addition to technical limitations, this study raises several ethical considerations related to bias, data usage, and real-world implications. Bias is a concern because it can occur at multiple stages of the machine learning process, including data collection, pre-processing, and model development [23]. The video game sales data may reflect existing industry biases, such as the dominance of certain platforms, publishers, or regions. For example, games from North America, Europe, and Japan may be overrepresented which could influence model predictions. If these biases are not addressed, machine learning models may reinforce these existing biases [23].

Another consideration involves data privacy and ethical data usage. While the dataset used in this project is publicly available and does not contain any personal information, it is still important to consider how data is collected and applied. While it would be more effect for predictive models in the gaming industry to incorporate user specific data such as player engagement metrics or purchase history, this would raise concerns about privacy and informed consent [24]. While these practices would improve model accuracy, they go against the guidelines and ethics of responsible data use, transparency, and accountability [25].

Finally, there are important real-world implications associated with the use of predictive models in the video game industry. Models like those developed in this study could influence decisions related to game development, marketing strategies, and platform selection . While this can improve efficiency and profitability, it may also lead to unintended consequences, such as prioritizing low-risk, high-profit games over innovative or niche titles [26]. This could reduce diversity in the gaming market and limit opportunities for smaller developers. Ethical guidelines highlight the need to consider fairness, societal impact, and long-term consequences when deploying machine learning models in real-world contexts [25].

8 Future Work and Recommendations

Future work for this study should focus on improving the dataset while carefully considering ethical implications. As mentioned previously, incorporating additional variables such as budgets, pricing and discount history, player engagement metrics, and exclusivity information could improve the machine learning models by capturing more factors that influence video game sales. However, these variables also introduce ethical concerns related to data privacy. In particular, variables such as player engagement and purchasing behavior may involve user-level data, which requires careful consideration of privacy, consent, and responsible data use [24][25]. Additionally, limiting or expanding the data set to recent game releases may also improve how well the model applies to new data. Future work will have to take care in balancing improved model accuracy with ethical data practices.

For future attempts at modeling, improvements can be made through better data preparation and feature engineering [21]. This could take the form of more advanced methods like deep learning models or optimized hyperparameters. By tuning these settings and selecting the most suitable models, predictive accuracy and stability can be further improved [22]. Additionally, alternative evaluation methods or additional performance metrics could provide a better evaluation of model performance [22].

When it comes to practical applications, predictive models such as those developed in this study can help make decision in the video game industry by forecasting sales performance. These models can help guide decisions within a company that are related to budgets, pricing, and platform selection by identifying patterns in the historical data [26]. For example, developers and publishers can use predictive analytics to learn player preferences, optimize release timing, and allocate their budgets better. However, reliance on these models may also encourage risk adverse decision-making that prioritizes safe and established trends over innovation [26].

9 Github

<https://github.com/eddie-li-git/CIND820>

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